

Where Does My Picture Go?

Maya and Pip explore data centers and keeping data safe.

A read-together book for ages 5-7.



A whale for Grandpa

Maya draws a whale and one bright star.

With Aunt Rae, she sends a copy to Grandpa.

“Where does my picture go?” asks Maya.

What picture would you like to send?

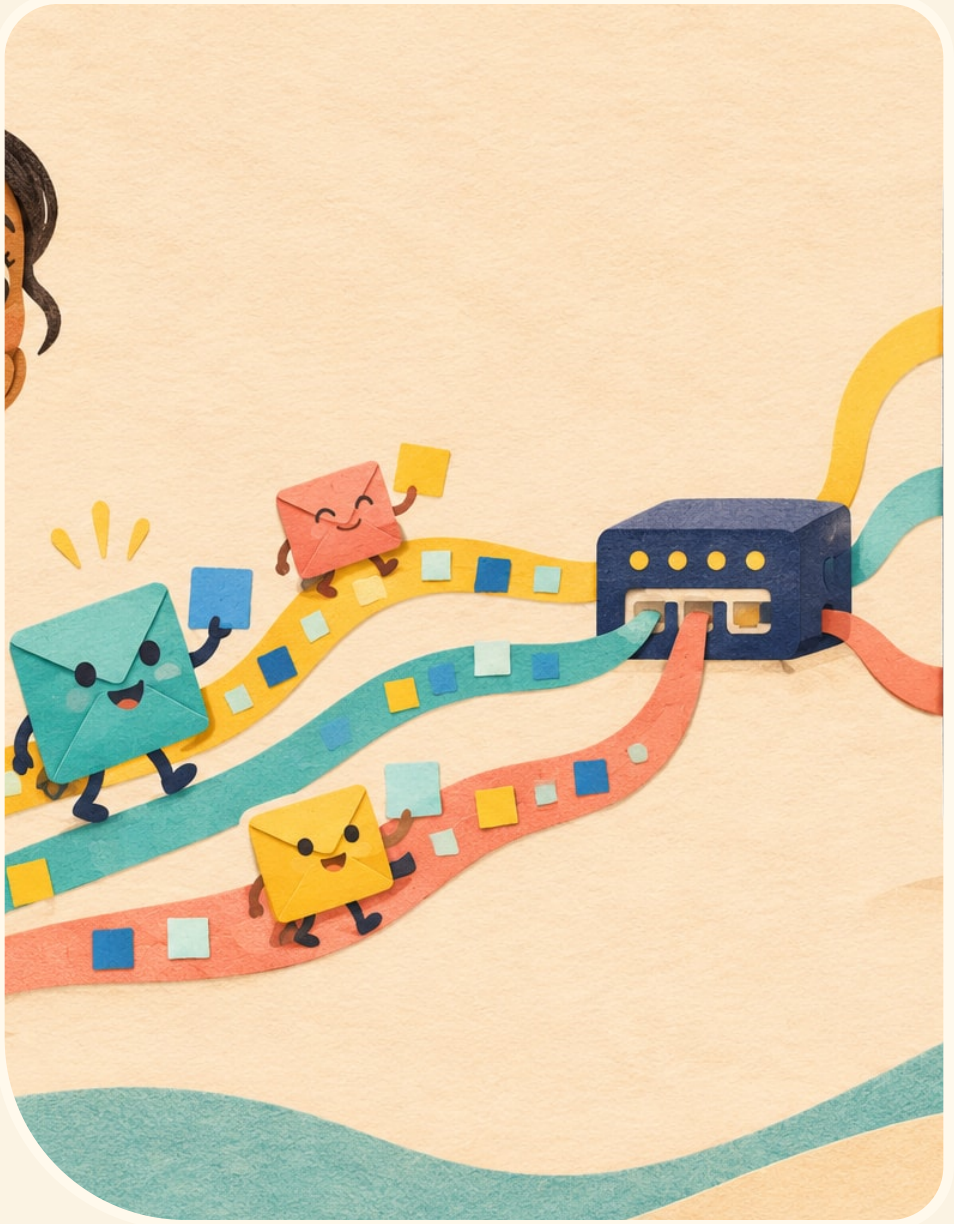
Meet Pip!

“Let’s imagine the trip,” says Rae.

Pop! Here is Pip, our pretend guide.

“Your picture is data,” says Pip. “Data means information a computer can use.”

Pictures, songs, and words can all be data. Can you name one more?





Try it with Pip

Trace three paths in the picture with your finger. Packets are pieces of data, not paper envelopes.

1 · A picture takes a trip

Little pieces, big trip

The picture travels in small pieces called packets.

They zip along network links.

At the other end, the pieces fit back together. Hello again, whale!

Count three pretend packets. Help them arrive.



1 · A picture takes a trip

The cloud is down here

“Is my picture in the sky?” asks Maya.

“No,” says Rae. “Cloud services use real computers.”

Some live in buildings called data centers. Let’s peek inside!

Point to the building that holds the computers.



2 · Inside the busy building

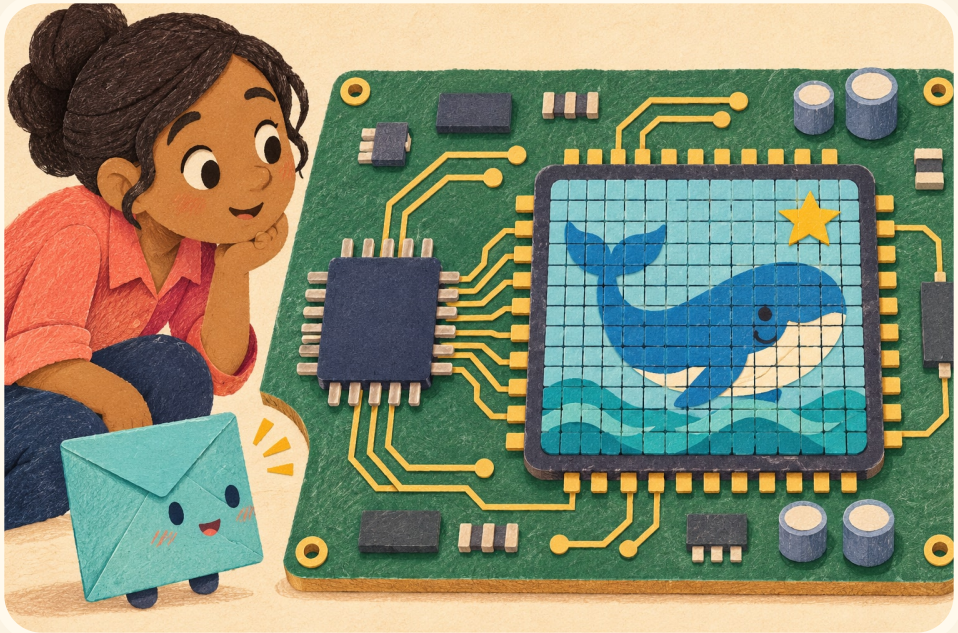
A room full of helpers

Hum, hum, hum.

Rows of computers work here. Many are called servers.

Some keep files. Some run apps. Some send the things we ask for.

The tall frames are called racks. Can you find one?



2 · Inside the busy building

Many little jobs at once

Next door, a computer makes a moving picture.

Its GPU chip does many small, similar jobs at once.

“Like a team filling lots of little squares!” says Maya.

Try it with Pip

Point to several squares at the same time with both hands. A GPU works on many similar calculations together.

Say the letters: G, P, U. Let’s light up the squares.

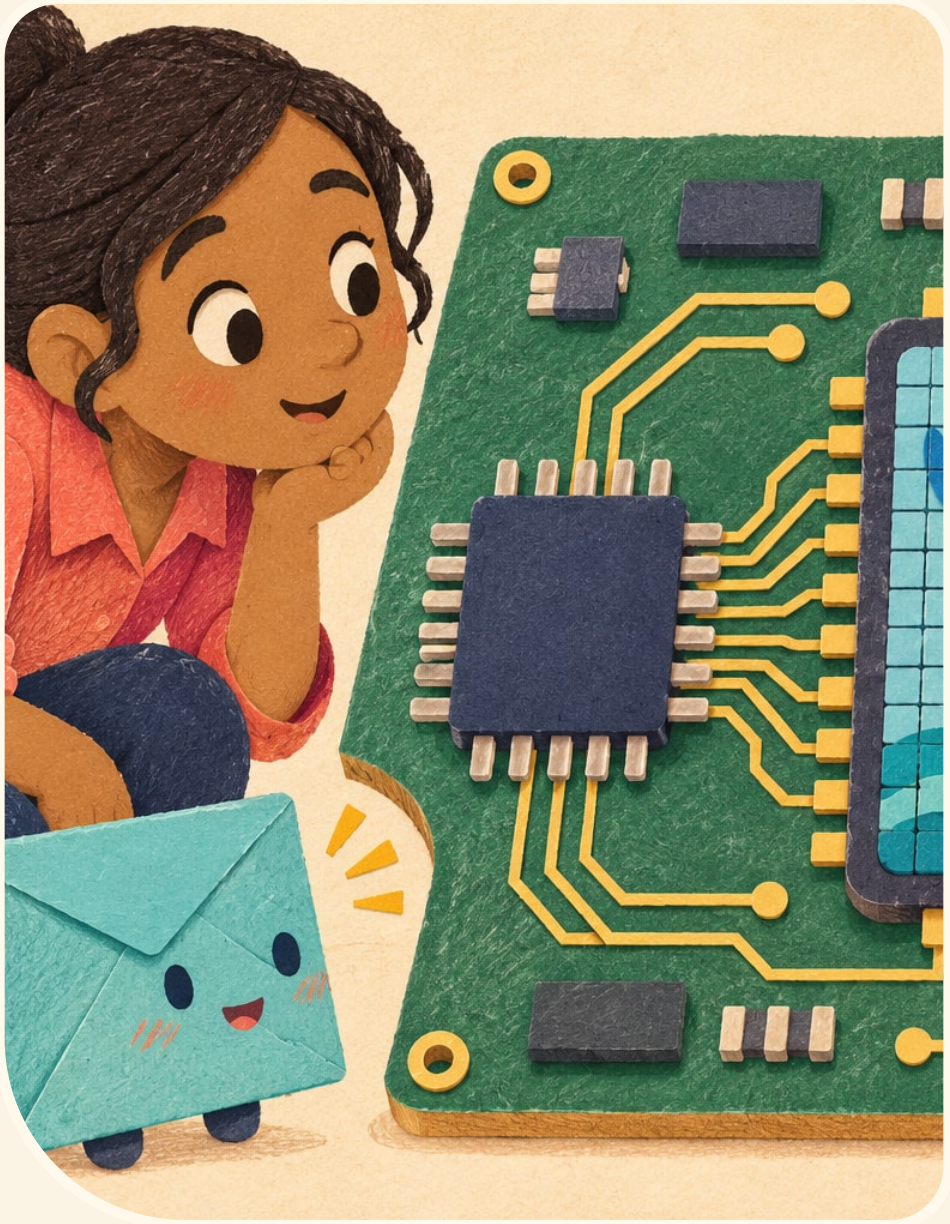
A tool that finds patterns

GPUs can help AI tools learn patterns from examples.

An AI tool might help find whales in pictures.

“It can get things wrong,” says Rae. “People must check.”

Could a cloud in a photo look like a whale?





2 · Inside the busy building

How is our computer doing?

Rae checks little clues called measurements.

How busy is the GPU? How warm? How much power is it using?

Is there room in its working memory?

Try it with Pip

Say these four clues together: busy, heat, power, room. A busy computer may be doing exactly the right job.

Explore the four things Rae checks.



A warm little clue

Beep! One computer is warmer than usual.

“Is someone being bad?” asks Pip.

“Not necessarily,” says Rae. “Let’s check its work and its cooling.”

When something is different, we look for clues before we decide.



2 · Inside the busy building

Whoosh goes the cool air

Rae finds a fan that needs fixing.

Fans move air to carry heat away. Some systems use liquid in special pipes.

The team fixes the fan. Whoosh!

Try it with Pip

Follow the blue air with your finger. Then follow the warm air out. Only trained grown-ups fix this equipment.

Help the pretend cooling team.



2 · Inside the busy building

Keep the lights on

Computers need electricity.

Backup batteries can help when power stops. Other power systems help too.

“We also try to waste less energy,” says Rae. “Our planet matters.”

What else in your home uses electricity?



Who may come in?

At the real building, workers check badges and permissions.

Visitors need help from a guide.

“Knowing who you are matters,” says Rae. “So does knowing where you may go.”

A badge is one check. It does not open every door.



Try it with Pip

Circle the allowed action: LOOK AT THE PICTURE / CHANGE THE PICTURE. Explain your choice.

3 · Keeping the picture safe

Look, but do not change

Files have permissions too.

Maya lets Grandpa see her picture.

That does not mean he may change it. People get just the access they need.

Grandpa has permission to look. What may he do?



3 · Keeping the picture safe

A special kind of lock

Encryption changes data so it is hard to read without the right digital key.

“Like a locked box?” asks Maya.

“That helps us imagine it,” says Rae.

Open and close our pretend picture box.

Try it with Pip

Make a box shape with your hands. Real encryption uses math, not a little metal lock.

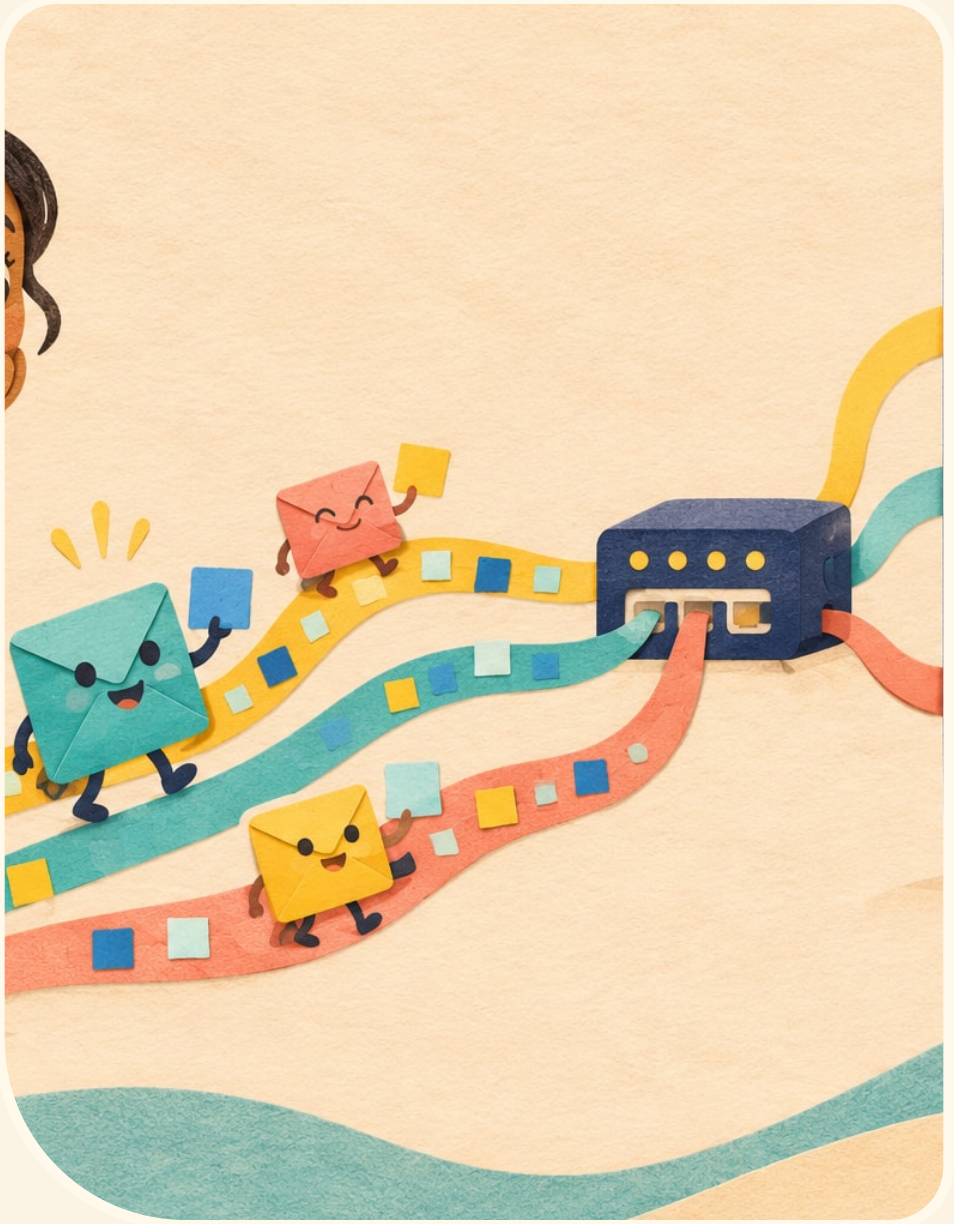
Paths with rules

Computers talk along network paths.

Rules help decide which computers may talk to which.

One open path does not mean every room is open.

Imagine a school: a library pass does not unlock every cupboard.





Try it with Pip

Choose: CHECK THE CLUES / DECIDE IT IS BAD AT ONCE. Why?

Knock, knock, knock

A computer sends many messages.

Beep! A tool spots a different pattern.

“A clue!” says Rae. “It might be a busy job, a mistake, or something unsafe.”

What should Rae do first?



Look a little closer

Rae checks the job list and the computer’s notes, called logs.

This time, an approved job is sending the messages.

“Busy can be okay,” says Pip.

Tell someone why a beep does not always mean danger.

This door stays closed

Next, a different request asks to open a private file.

It has no permission. The system says no.

The security team checks what happened and keeps a record.

People and tools work together. No one lock does every job.



Where did the star go?



Now Pip spots a problem.

In one saved copy, Maya’s star has become a circle!

Was it a mistake? An unwanted change? The team needs more clues.

Find the part of the picture that changed.

Try it with Pip

Point to the star and the circle. A change tells us to check; it does not tell us who made it.



Does it still match?

Computers can check files against a trusted record.

A check can show that something changed.

“It cannot tell us the whole story,” says Rae. “We still investigate.”

Keeping data from unwanted changes is called integrity. That is a big word for an important job!



3 · Keeping the picture safe

A copy kept safe

Good news! The team kept a backup.

A backup is an extra copy stored safely, apart from the first.

They check it before using it.

Find the extra copy in the other cabinet.



Hello again, little star

The team brings back the checked copy.

There is Maya’s star!

They also look for what caused the change, so they can help stop it happening again.

Try it with Pip

Draw a star in the air. Restoring a copy helps us recover; finding the cause is another job.

Bring back the picture from our pretend backup.



4 · Home again

A whale-sized smile

Later, Maya visits Grandpa. Her picture is on his screen.

“It took a whole team!” says Maya.

Grandpa gives her a whale-sized hug. Pip waves. What a trip!

The end. What helped the picture get there safely?



Your turn

You can ask for help

If a message feels odd, pause and ask a trusted grown-up.

Ask before sharing a photo or private information.

If you clicked by mistake, tell someone. You can get help.

Try it with Pip

Name a trusted grown-up you could ask. You never have to solve a scary online problem alone.

A pop-up asks for your home address. What do you do?

Little word garden

New words to grow

Say a word. Point to something in the story that matches it.

Data

Information a computer can use.

Server

A computer that does jobs for other computers.

Network

Connections that let computers share data.

GPU

A chip that does many similar jobs at once.

Permission

What someone is allowed to do.

Backup

An extra copy kept safely.

Integrity

Keeping data from unwanted changes.

Security

Ways to help protect people, computers, and data.



Play together · 1

Tell the picture’s journey

Put these story stops in order.

Start with Maya. Finish with Grandpa.

Tell the story in your own words.

Tap each stop in the order it belongs.

Try it with Pip

Write 1, 2, 3 beside these stops: GRANDPA’S SCREEN __ / MAYA’S TABLET __ / DATA CENTER __. Draw a line to join them.



Play together · 2

Which job is allowed?

Pretend you care for the picture.

Grandpa may look. Rae may check the backup.

Someone else asks to erase it. They have no permission.

Which request should stay closed?

Try it with Pip

Circle the request to refuse: GRANDPA LOOKS / RAE CHECKS / SOMEONE ERASES WITHOUT PERMISSION.

Make your own safe journey

On paper, draw a picture you would like to send.

Add a path, a computer, a cooling fan, and a safe backup.

Tell a grown-up what each part does.

What will your picture be?

Use the space below. Your picture does not need your name, address, or any private information.

For grown-ups

Read, wonder, check

Read a few pages at a time. Invite pointing, sounds, and questions. Ages 5-7 do not need to memorize the technical words. The printable activities work without a device.

Pip, the journey, and its problems are fictional. Envelopes, cabinets, locks, and colored tiles are visual metaphors. Packets are data; GPUs do calculations. Not every photo transfer needs a GPU or visits the same kinds of systems.

The activities are pretend examples, not live monitoring, encryption, or access control. They run in the browser and do not ask for personal information. Read-aloud uses a device/browser voice if available; voice processing depends on that provider.

Busy GPUs and unusual traffic do not prove an attack. A changed cryptographic hash indicates a mismatch with a trusted reference; it does not identify the person, intent, or correctness of the original file. Backups must be protected and tested.

Children can pause and ask for help. Adults handle accounts, updates, strong credentials, and extra sign-in checks. Privacy, permission, integrity, and recovery all matter. No single tool guarantees safety.

Printing: 32 landscape pages. Choose Fit on Letter or A4 paper. For a folded booklet, use your printer’s booklet setting. Read-aloud depends on browser/device support; no recording is collected by this book.

The real ideas: data centers

The story simplifies these primary references. Read together with an adult. Sources checked November 5, 2026.

AWS · What is a data center?

Servers, storage, networks, cooling, power, and cloud infrastructure; pages 5-6 and 10-12.

https://aws.amazon.com/what-is/data-center/

NVIDIA · CPU and GPU

Many similar calculations in parallel; pages 7-8. The colored tiles are a teaching metaphor.

https://blogs.nvidia.com/blog/whats-the-difference-between-a-cpu-and-a-gpu/

NVIDIA · DCGM Exporter

GPU monitoring and measurements; page 9. Our activities do not read real GPU measurements.

https://docs.nvidia.com/datacenter/cloud-native/gpu-telemetry/latest/dcgm-exporter.html

AWS · Data center controls

Physical access, permissions, logging, and layers of protection; pages 13-14 and 19.

https://aws.amazon.com/trust-center/data-center/our-controls/

The real ideas: keeping data safe

Use these references for the details behind our simple explanations. The story describes general ideas, not a deployed cloud system.

NIST · Encryption

Transforming readable data using cryptography; page 15. A screen cover is not real encryption.

https://csrc.nist.gov/glossary/term/encryption

NIST · Cryptographic hash function

Hash functions and change detection; pages 20-21. Reference values also need protection.

https://csrc.nist.gov/glossary/term/cryptographic_hash_function

NCSC · Backing up your data

Separate safe copies and restoring files; pages 22-23.

https://www.ncsc.gov.uk/collection/top-tips-for-staying-secure-online/always-back-up-your-most-important-data

FTC · Recognize and avoid phishing scams

Unexpected requests and checking before responding; page 25. An adult should help young children.

https://consumer.ftc.gov/articles/how-recognize-avoid-phishing-scams